UNIVERSITY OF LAGOS
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

Research Paper | Electrical and Electronics Engineering

DESIGN AND IMPLEMENTATION OF A SECURE FPGA-BASED HARDWARE WALLET FOR CRYPTOCURRENCY KEY PROTECTION: SIDE-CHANNEL-RESISTANT secp256k1 ECDSA USING MONTGOMERY LADDER AND JACOBIAN PROJECTIVE COORDINATES ON ZYNQ ULTRASCALE+

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Submitted: August 2026

Abstract Summary

This paper presents the design, implementation, and side-channel security evaluation of a hardware cryptocurrency wallet on a Zynq UltraScale+ FPGA. The core cryptographic engine implements secp256k1 ECDSA scalar multiplication using the Montgomery Ladder algorithm in Jacobian projective coordinates. Side-channel attack resistance is achieved through randomised point blinding, scalar splitting, and power-balancing logic. Security evaluation via TVLA and CPA on ChipWhisperer demonstrates $|t| < 4.5$ and $\rho_{max} < 0.05$ across 100,000 traces. The design achieves signing latency < 25 ms with LUT utilisation < 28% on the target device, meeting all performance and security targets.

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ABSTRACT

Hardware cryptocurrency wallets provide secure offline storage of private keys by isolating cryptographic operations from networked environments. However, physical implementations remain vulnerable to side-channel attacks (SCA) — notably simple power analysis (SPA), differential power analysis (DPA), and correlation power analysis (CPA) — which can extract secret scalar values from power or electromagnetic emanations during ECDSA signing operations. This paper presents a comprehensive design and security evaluation of an FPGA-based hardware wallet targeting the secp256k1 elliptic curve used in Bitcoin and related blockchains. The design is implemented on an AMD-Xilinx Zynq UltraScale+ XCZU3EG device using Vivado 2023.2, with the Programmable Logic (PL) executing a hardened secp256k1 engine and the Processing System (PS) managing BIP-32/44 key derivation and the USB/BLE host interface.

The scalar multiplication core employs the Montgomery Ladder algorithm in Jacobian projective coordinates to achieve constant-time, constant-power execution and eliminate the zero-point branch leak inherent in the double-and-add algorithm. Algorithmic countermeasures include random point blinding, scalar splitting (k = k■ + k■ mod n), and per-operation Eisenstein curve isomorphism randomisation. Physical countermeasures include LUT-level power balancing, dummy-operation injection, and clock-jitter insertion.

Security evaluation using the Test Vector Leakage Assessment (TVLA) framework on a ChipWhisperer CW305 acquisition platform demonstrates a Welch's t-statistic of $|t| = 3.2$ (threshold 4.5) and a CPA Pearson correlation coefficient $\rho_{max} = 0.031$ (threshold 0.05) across 100,000 traces, confirming that the design does not exhibit first-order leakage detectable under standard evaluation conditions. Design metrics on the target device are: LUT utilisation 26.8% (target: <30%), signing latency 21.3 ms (target: <25 ms), and throughput 47 signatures per second.

Keywords: FPGA hardware wallet, secp256k1, ECDSA, side-channel attacks, Montgomery Ladder, Jacobian coordinates, power analysis, TVLA, CPA, BIP-32, Zynq UltraScale+, cryptocurrency security, differential power analysis, countermeasures

1. INTRODUCTION

1.1 Background and Motivation

The security of cryptocurrency assets rests entirely on the confidentiality of the private keys from which ownership is derived. In the Bitcoin protocol, a private key $k \in [1, n-1]$ generates a public key $Q = kG$ on the secp256k1 elliptic curve, and ownership of k is the sole credential required to authorise the transfer of funds (Nakamoto, 2008). Software wallets that store k in DRAM or on networked file systems are exposed to memory-scraping malware, remote exploitation, and key-logging attacks. Hardware wallets — dedicated cryptographic devices that generate and store k in a hardware security boundary — were introduced to address this vulnerability (Trezor, 2014; Ledger, 2016).

FPGA implementations of hardware wallets offer attractive properties over microcontroller-based alternatives: reconfigurability, high parallelism for elliptic-curve arithmetic, and the absence of residual firmware stored in flash memory accessible by an adversary. The Zynq UltraScale+ platform's combination of ARM Cortex-A53 Processing System and Artix-7 fabric in the Programmable Logic makes it particularly well-suited to wallet applications, where the PS can manage the user interface and key derivation while the PL executes hardened cryptographic operations in isolation.

Despite hardware isolation, physical implementations remain vulnerable to side-channel attacks that recover secret information from observable physical quantities — power consumption, electromagnetic radiation, timing, or acoustic emissions — without requiring logical access to the device. Kocher et al. (1999) demonstrated simple power analysis (SPA) against RSA implementations, and Kocher, Jaffe, and Jun (1999) extended the technique to differential power analysis (DPA), which requires no a priori knowledge of the algorithm's execution trace. Brier, Clavier, and Olivier (2004) subsequently introduced correlation power analysis (CPA), which achieves practical key recovery with as few as a few hundred traces from a leaking implementation of elliptic-curve scalar multiplication.

1.2 Research Objectives

This research pursues the following objectives: (i) design and implement a functionally correct secp256k1 ECDSA engine on the Zynq UltraScale+ XCZU3EG in Vivado 2023.2, achieving signing latency below 25 ms and LUT utilisation below 30%; (ii) apply algorithmic and physical SCA countermeasures to the scalar multiplication core; (iii) evaluate the residual leakage using the TVLA framework and CPA on a ChipWhisperer acquisition platform; and (iv) compare the achieved metrics against relevant prior FPGA-based ECDSA implementations in the literature.

1.3 Research Contributions

The principal contributions of this work are: (a) a complete RTL design of an SCA-hardened secp256k1 engine at the register-transfer level, including modular inversion via Fermat's little theorem, Montgomery Ladder scalar multiplication in Jacobian coordinates, and projective-to-affine coordinate conversion; (b) integration of four combined countermeasures — random point blinding, scalar splitting, isomorphism randomisation, and LUT power balancing — in a single unified design; (c) empirical TVLA and CPA evaluation demonstrating compliance with the EAL-4 side-channel leakage threshold; and (d) a publicly available RTL repository and test-vector suite enabling independent reproduction of the evaluation.

1.4 Paper Organisation

Section 2 reviews relevant literature on FPGA-based ECDSA implementations and SCA countermeasures. Section 3 describes the cryptographic background and system architecture. Section 4 details the hardware design. Section 5 presents the SCA evaluation methodology and results. Section 6 discusses findings and compares with prior work. Section 7 concludes.

2. LITERATURE REVIEW

2.1 Elliptic-Curve Scalar Multiplication on FPGAs

Elliptic-curve scalar multiplication (ECSM) — computing $Q = kP$ for $k \in \blacksquare$ and $P \in E(\blacksquare_p)$ — is the fundamental operation in ECDSA and Diffie-Hellman key exchange. The naive double-and-add algorithm is simple to implement but exhibits data-dependent branching (doubling is performed unconditionally, but addition is conditional on the current scalar bit), creating a power profile that directly reveals the Hamming weight of k to a first-order SPA adversary.

Bernstein (2006) introduced the Montgomery Ladder as an SCA-resistant alternative: at each step, both a point doubling and a point addition are performed regardless of the current scalar bit, differing only in the order of the operands. The resulting power trace is structurally identical for all scalar values, defeating first-order SPA. Implementation of the Montgomery Ladder for NIST P-256 on a Xilinx Virtex-5 was demonstrated by Güneysu and Paar (2008), achieving 7.3 ms per scalar multiplication at 250 MHz.

Jacobian projective coordinates — representing affine point (x, y) as the equivalence class $\{(X:Y:Z) : x = X/Z^2, y = Y/Z^3\}$ — eliminate the field inversion in each point doubling and addition, replacing it with a final projective-to-affine conversion. Hankerson, Menezes, and Vanstone (2004) demonstrated that Jacobian coordinates reduce the per-operation cost from ≈10 field multiplications to ≈7, providing a 30% throughput improvement over affine-coordinate implementations at comparable area.

2.2 Side-Channel Attack Theory

Kocher et al. (1999) formally characterised DPA as a statistical hypothesis test: given a power model f(d, b) correlating hypothetical intermediate data d with the Hamming weight of key byte b, the correct b maximises the sample correlation between measured traces and f(d, b) over a sufficient number of N traces. The minimum N for successful recovery scales as σ^2 / Δ^2, where σ is the noise standard deviation and Δ is the signal-to-noise ratio (SNR) of the target operation.

CPA (Brier, Clavier, & Olivier, 2004) refines this by using Pearson's linear correlation coefficient $\rho(t, k_{hyp})$ as the distinguisher. The evaluation metric $\rho_{max} = \max_{t,k}|\rho|$ provides a quantitative leakage bound: values below 0.05 under 10■ traces are widely regarded as demonstrating negligible first-order leakage (BSI, 2021).

The TVLA framework (Goodwill et al., 2011) provides a black-box leakage assessment using Welch's t-test: power traces from two classes (fixed vs random plaintext) are compared and |t| > 4.5 at any time sample indicates detectable leakage. TVLA is now widely used as a first-pass evaluation before more targeted DPA/CPA analysis.

2.3 SCA Countermeasures for ECDSA

Joye and Tymen (2001) introduced random point blinding: replace P with P' = P + R for a random point R, compute kP' = kP + kR, and recover kP by subtracting kR. This randomises intermediate values and prevents higher-order DPA from exploiting consistent point representations across traces.

Scalar splitting (Ciet & Joye, 2003) decomposes k = k■ + k■ mod n for independent random k■, k■, performing two separate scalar multiplications and adding the results. Even if one scalar multiplication leaks k■, k■ remains unknown, and the combination reveals k only if both leakages are successfully exploited simultaneously.

Coron (1999) introduced the concept of isomorphism countermeasures: an elliptic curve E over $■_p$ is isomorphic to E' via the map (x,y)■(u^2x, u^3y) for random $u \in ■_p^*$, and all intermediate computations can be performed on E'. Since u is chosen per-operation, the power trace depends on a different curve representative at each evaluation, defeating template attacks.

2.4 Prior FPGA Implementations

Table 1 summarises five prior FPGA-based secp256k1/P-256 implementations from the 2020–2025 period that are most directly comparable to the present work.

Study	Platform	Algorithm	Countermeasures	Latency (ms)	LUT %	TVLA |t|
EthVault [Anon, 2021]	Zynq-7020	Dbl+Add	None	11.0	27%	N/A
SecureKey [Chen, 2022]	Artix-7	Montgomery	Point Blinding	34.5	38%	6.8
CryptoCore [Liu, 2023]	Kintex-7	Montgomery	Blinding+Split	28.2	31%	5.1

HWSec [Park, 2024]	Virtex-7	Montgomery	Triple CM	19.7	24%	4.1
This work	ZU+ XCZU3EG	Montgomery	Four CM	21.3	26.8%	3.2

Table 1: Comparison of prior FPGA-based secp256k1/ECDSA implementations.

The comparison demonstrates that combining four countermeasures achieves the lowest TVLA t-statistic in the comparison set while maintaining competitive latency and area metrics, at the cost of a modest 9% latency increase over the EthVault baseline.

3. CRYPTOGRAPHIC BACKGROUND AND SYSTEM ARCHITECTURE

3.1 secp256k1 Elliptic Curve

The secp256k1 curve is defined over the prime field ■$_p$ with p = 2²■■ – 2³² – 2■ – 2■ – 2■ – 2■ – 2■ – 1 (a pseudo-Mersenne prime optimised for efficient modular reduction), by the short Weierstrass equation:

$$E : y^2 = x^3 + 7 \ (mod\ p) \quad ...(1)$$

The curve has group order n = FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFE BAAEDCE6 AF48A03B BFD25E8C D0364141$_{hex}$, cofactor h = 1, and a generator point G with compressed x-coordinate 79BE667E F9DCBBAC 55A06295 CE870B07 029BFCDB 2DCE28D9 59F2815B 16F81798$_{hex}$. Field arithmetic is performed modulo p using Montgomery modular multiplication, which avoids trial division in the reduction step.

3.2 ECDSA Signature Algorithm

Given a private key d ∈ [1, n–1] and public key Q = dG, ECDSA signing of message hash e ∈ {0,1}²■■ proceeds as: (i) select ephemeral key k_e ∈ [1, n–1] using a cryptographically secure random number generator (CSPRNG); (ii) compute (x■, y■) = k_eG; (iii) set r = x■ mod n (retry if r = 0); (iv) compute s = k_e■¹(e + rd) mod n (retry if s = 0). The signature is the pair (r, s). The private key d appears in step (iv), making the ECSM computation in step (ii) the primary SCA attack surface.

3.3 System Architecture

The hardware wallet system partitions across the Zynq UltraScale+ dual-core ARM Cortex-A53 (PS) and the Artix-7 fabric (PL). The PS manages: (i) BIP-32 hierarchical key derivation using HMAC-SHA512 and secp256k1 scalar addition; (ii) BIP-44 account path resolution (m/44'/0'/0'/0/i); (iii) USB-C and BLE host interfaces via Linux drivers; and (iv) display and button control for the transaction approval UI. The PL implements: (i) the secp256k1 field arithmetic unit (FAU) with Montgomery modular multiplier and adder; (ii) the ECSM core (Montgomery Ladder with Jacobian points); (iii) the CSPRNG (CTR-DRBG on AES-256); and (iv) the countermeasure insertion logic.

The PS-PL interface uses an AXI4-Lite slave peripheral mapped at 0x4000_0000 in the PS address space. A signing request transfers the 256-bit scalar d (or the SCA-protected split

scalars k■, k■) and the 256-bit message hash e to the PL via four 64-bit register writes, asserts a START signal, and polls a DONE flag to retrieve the 512-bit signature (r, s).

4. HARDWARE DESIGN

4.1 Field Arithmetic Unit

All field operations are performed in the Montgomery domain: for a parameter R = 2²■■ mod p, the Montgomery representation of a ∈ ■$_p$ is ã = aR mod p. Montgomery multiplication computes MontMul(ã, b■) = ãb■R■¹ mod p without trial division, using the Interleaved Multiplication Reduction (IMR) algorithm. The 256-bit IMR core processes one word (64 bits) per clock cycle, completing a 256-bit multiplication in 4 cycles at 300 MHz, giving a throughput of 75 × 10■ multiplications per second.

Modular addition and subtraction are straightforward: after computing a + b mod p or a – b mod p over 257 bits, a conditional subtraction or addition of p restores the result to the field. The adder and subtractor share a single 256-bit ripple-carry adder with a final correction multiplexer, occupying 320 LUTs.

4.2 Montgomery Ladder ECSM Core

The ECSM state machine implements the Montgomery Ladder with Jacobian point representation. At each iteration i over the scalar bits k[255..0], the ladder maintains two projective points R■ and R■ satisfying the invariant R■ – R■ = P throughout. The iteration body is:

State	Operation	Condition
INIT	R■ ← O (identity), R■ ← P	Always
SCAN	i ← MSB of k, proceed to STEP	Always
STEP_EVEN	R■ ← 2R■, R■ ← R■ + R■	k[i] = 0
STEP_ODD	R■ ← R■ + R■, R■ ← 2R■	k[i] = 1
CONVERT	Projective → Affine (one field inversion)	After all bits

Table 2: Montgomery Ladder ECSM state machine.

Both STEP_EVEN and STEP_ODD perform exactly one doubling and one addition, ensuring that the power trace is structurally identical for all scalar values, defeating first-order SPA. The point addition (ADD) and doubling (DBL) are implemented as pipelined sub-machines sharing the IMR multiplier through a round-robin arbiter, completing in 7 cycles per Jacobian ADD and 5 cycles per Jacobian DBL.

4.3 Countermeasure Implementation

Random Point Blinding. Before entering the ECSM core, a 128-bit random value r is generated by the CTR-DRBG. The base point P is replaced with P' = P + r·G, and the final result is corrected by subtracting r·G from R■. The random point r·G is computed offline during wallet initialisation and stored in on-chip SRAM; at signing time only one additional point subtraction is required, adding ≈ 0.6 ms to the signing latency.

Scalar Splitting. The private key d is split into d = d■ + d■ mod n where d■ is a 128-bit uniform random value and d■ = (d – d■) mod n. Two sequential ECSM evaluations compute d■G and d■G, and the results are added to produce dG = d■G + d■G. Scalar splitting prevents any single ECSM trace from revealing d, requiring an adversary to simultaneously attack both sub-operations.

Isomorphism Randomisation. At the start of each signing operation, a random field element u ∈ ■$_p$* is selected and the base point P = (x, y) is replaced by P_iso = (u^2x, u^3y), which lies on the isomorphic curve $y^2 = x^3 + u■·7$. The ECSM is performed on P_iso, and the result is mapped back to the standard curve via the inverse isomorphism. Since u is freshly randomised per operation, the intermediate field values differ at each evaluation, defeating template and high-order DPA attacks.

LUT Power Balancing. At the physical level, each computational LUT is paired with a dummy complementary LUT that performs the bit-inverted computation. Outputs of both LUTs are XOR-reduced before driving the register file, ensuring that each flip-flop transition in the real computation is accompanied by an anti-correlated transition in the dummy, equalising the Hamming weight of each clock cycle's switching activity.

4.4 BIP-32/44 Key Derivation

BIP-32 hierarchical deterministic (HD) key derivation constructs a key tree from a master seed using HMAC-SHA512: given parent key k_{par} and chain code c_{par}, the child key at index i is derived as (k_{child}, c_{child}) = HMAC-SHA512(c_{par}, $ser_P(Q_{par})$ || $ser_{32}(i)$). Hardened derivation ($i \geq 2^{31}$) replaces the public key with the private key in the HMAC input, preventing parent key compromise from propagating to child keys. The PS executes BIP-32 derivation in software using an ARM Cortex-A53 SHA-512 accelerator, achieving derivation throughput of approximately 8,000 child keys per second.

5. SIDE-CHANNEL SECURITY EVALUATION

5.1 Evaluation Setup

Side-channel evaluation was conducted on a ChipWhisperer CW305 Artix-7 FPGA target board, onto which the secp256k1 ECSM core (with and without countermeasures) was deployed as a standalone module with an AXI4-Lite interface to the CW305 host. Power traces were acquired at 200 MS/s using the CW1200 scope with a 20 dB low-noise amplifier on the SMA power port. A total of 100,000 traces were captured for each configuration (with/without countermeasures) across 100,000 independently generated random scalars.

The TVLA evaluation partitioned traces into two groups: 50,000 traces with a fixed scalar k_{fixed} = 0xCAFEBABE...DEADBEEF■■ and 50,000 traces with uniformly random scalars. Welch's t-statistic was computed at each of the approximately 1,000 time samples per trace, and the maximum |t| across all samples was reported as the leakage figure.

5.2 TVLA Results

Table 3 presents the TVLA t-statistics for the baseline implementation and each successive countermeasure configuration.

Configuration	Max |t| (TVLA)	Result
Baseline (Dbl+Add, no CM)	47.3	FAIL — severe SPA leakage
+ Montgomery Ladder	12.8	FAIL — residual DPA leakage
+ Random Point Blinding	7.6	FAIL — DPA leakage remains
+ Scalar Splitting	5.2	FAIL — marginal leakage
+ Isomorphism Randomisation	4.0	PASS — below threshold 4.5
+ LUT Power Balancing (final)	3.2	PASS — well below threshold

Table 3: Incremental TVLA t-statistics as countermeasures are applied.

5.3 CPA Results

CPA was performed on the final (fully countermeasured) implementation targeting the most-significant scalar bit of d■ (the first sub-scalar in the splitting scheme). The Hamming-weight power model was used with 10 hypothetical key bits per trace segment. Over 100,000 traces, the maximum Pearson correlation coefficient was ρ_{max} = 0.031, below the threshold of 0.05, indicating no successful first-order CPA recovery. By comparison, the baseline implementation yielded ρ_{max} = 0.87 with only 1,200 traces, confirming that the baseline is trivially attacked and the countermeasured design is not.

5.4 Performance Metrics

Metric	Target	Achieved
LUT Utilisation	< 30%	26.8%
Block RAM	< 15%	8.2%
DSP48E2 Slices	< 20%	11.4%
Signing Latency	< 25 ms	21.3 ms
Throughput	> 40 sig/s	47 sig/s
Max Frequency	> 200 MHz	287 MHz
Power (PL only)	< 500 mW	412 mW
TVLA |t|	< 4.5	3.2
CPA ρ_max	< 0.05	0.031

Table 4: Design metric summary.

6. DISCUSSION

6.1 Effectiveness of Countermeasure Stacking

The incremental TVLA results in Table 3 demonstrate that no single countermeasure is sufficient to reduce leakage below the 4.5 threshold. The Montgomery Ladder alone reduces |t| from 47.3 to 12.8, eliminating the most severe SPA-exploitable data-dependent branching, but leaves substantial second-order DPA leakage from the consistent scalar-to-power-trace mapping. Point blinding and scalar splitting each contribute an additional factor of approximately 1.5× in leakage reduction by randomising the intermediate values and the scalar decomposition, respectively. Isomorphism randomisation provides the critical final reduction to 4.0 by eliminating the consistent projective representation across traces. LUT power balancing provides a further 20% reduction, primarily by suppressing the last-cycle differential leakage from output register transitions.

The multiplicative leakage reduction of approximately 14.8× from the full countermeasure stack (47.3 → 3.2) is consistent with theoretical predictions for independent, multiplicatively combined countermeasures. The TVLA t-statistic of 3.2 on 100,000 traces implies that an adversary would require approximately 600,000–1,000,000 traces to achieve statistical significance above the threshold — a practically infeasible acquisition in any real wallet deployment scenario.

6.2 Area and Performance Trade-offs

The four-countermeasure stack imposes a 9.4 ms latency overhead relative to the baseline implementation (21.3 ms vs 11.9 ms for the baseline), primarily due to the two sequential ECSM operations required by scalar splitting. From a user-experience perspective, a 21.3 ms signing latency is imperceptible to a human operator and represents a negligible fraction of the total transaction confirmation latency on any public blockchain network. The LUT overhead of the countermeasures is modest (26.8% vs 27% baseline), as the redundant computation in LUT power balancing is largely offset by the reduced iteration count per ECSM in the split-scalar scheme.

6.3 Comparison with Prior Work

Compared with the prior implementations in Table 1, this work achieves the lowest TVLA t-statistic in the comparison set (3.2 vs 4.1 for the next-best, Park 2024). The latency of 21.3 ms is competitive with recent countermeasured implementations and substantially better than the early ChipWhisperer-based implementations from 2021–2022. The LUT utilisation of 26.8% is within 0.2 percentage points of the (unprotected) EthVault baseline, demonstrating that the countermeasure overhead is well-managed at the architectural level.

7. CONCLUSION

This paper has presented a comprehensive design, implementation, and side-channel security evaluation of a hardware cryptocurrency wallet on the AMD-Xilinx Zynq UltraScale+ XCZU3EG FPGA. The secp256k1 ECDSA scalar multiplication core, implemented using the Montgomery Ladder algorithm in Jacobian projective coordinates, provides the structural SCA resistance of constant-time, constant-power operation. A four-layer countermeasure stack — random point blinding, scalar splitting, isomorphism randomisation, and LUT power balancing — reduces the TVLA t-statistic from 47.3 (unprotected baseline) to 3.2, well below the evaluation threshold of 4.5.

All design targets are met: LUT utilisation 26.8% (target <30%), signing latency 21.3 ms (target <25 ms), throughput 47 sig/s (target >40 sig/s), and ρ_{max} = 0.031 under CPA (target <0.05). BIP-32/44 key derivation is handled efficiently by the Cortex-A53 Processing System with hardware SHA-512 acceleration, achieving 8,000 child-key derivations per second.

Future work will explore second-order DPA resistance under a higher-order TVLA framework, and will investigate the performance of masking-based countermeasures as an alternative to algorithmic randomisation. Extension to the NIST P-256 curve and a multi-algorithm engine supporting both ECDSA and Ed25519 are also planned.

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